

GENETICS 101 · RECITATION HOMEWORK

Homework 1: The Instruction Book

Name: _____

Date: _____

Instructions: About 20 minutes. Part A is a decoding puzzle — read the code table carefully, three letters at a time. Total: **21 points**.

Part A — Decode the message

Cells read DNA three letters at a time. Here is a pretend code table: ATG = B, CCA = A, GGT = Y, TTA = O, GAC = L, CTG = E. Message to decode: ATG CCA ATG GGT

1. Decode the message above, one three-letter word at a time. Write the word it spells. (4 pts)

Part B — Counting the book

2. A baby receives 23 chromosomes from its mother and 23 from its father. How many chromosomes does the baby's body cell have? (1 pt)

3. Before a body cell divides, it copies every one of its chromosomes. How many chromosomes are inside it just before it splits? (1 pt)

4. DNA has 4 letters, and cells read them in 3-letter words. How many different 3-letter words are possible? (Hint: 4 choices for the first letter × 4 for the second × 4 for the third.) (2 pts)

Part C — Identical or fraternal?

5. Write IDENTICAL or FRATERNAL for each pair of twins. (4 pts)

Twins	Identical or fraternal?
Two babies who grew from one egg that split in two	
A boy and a girl born together	
Two babies who grew from two separate eggs	
Two sisters who look exactly alike and share the same DNA	

Part D — Your family tree

6. Draw a family tree: you at the bottom, your two parents above you, and your four grandparents above them. Draw an arrow from each parent to you and write “23” on each arrow to show the chromosomes you received. (2 pts)

7. What did Rosalind Franklin's “Photo 51” show? (1 pt)

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- A. The X-shaped pattern that revealed DNA is a twisted ladder (a double helix).
- B. The first photograph of a chromosome under a microscope.
- C. The four DNA letters lined up in order along one gene.
- D. Two identical twins' DNA compared side by side.

Part E — Flashback

8. From Biology: which part of the cell holds the DNA? (1 pt)

9. From NICU Tutor and Chemistry: hemoglobin is a _____ that carries _____. (2 pts)

Part F — Think like a doctor

10. Twins fill a lot of NICU beds. Give TWO reasons twins are more likely than single babies to need the NICU, and say which NICU Tutor machine each twin might need first. (3 pts)

Score: _____ / 21 points

Professor's comment:
